

SKILL – 9:

PART 1: Directory Navigation

Code:

```
GNU nano 2.9.2 skill9_navigation.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>

int main() {
    char current[PATH_MAX];
    char previous[PATH_MAX] = "";
    char path[PATH_MAX];

    if (getcwd(current, sizeof(current)) == NULL) {
        perror("getcwd");
        return 1;
    }

    printf("=== Directory Navigation ===\n");
    printf("Current Directory: %s\n", current);

    while (1) {
        printf("\nEnter directory path:");
        printf("(or 'prev' to go back, 'pwd' to display, 'exit' to quit): ");

        if (fgets(path, sizeof(path), stdin) == NULL)
            break;

        path[strcspn(path, "\n")] = '\0';

        if (strcmp(path, "exit") == 0) {
            printf("Exiting navigation...\n");
            break;
        }

        if (strcmp(path, "pwd") == 0) {
            if (getcwd(current, sizeof(current)) != NULL)
                printf("Current Directory: %s\n", current);
            continue;
        }

        if (strcmp(path, "prev") == 0) {
            if (previous[0] != '\0')
                strcpy(current, previous);
            else
                return 1;
        } else {
            if (chdir(path) != 0)
                perror("chdir");
            else
                strcpy(previous, current);
        }
    }
}
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-09$ ./skill9_navigation
=== Directory Navigation ===
Current Directory: /home/bhardwaj/2520090182_OSSP/2520090182_Skill/Skill-09

Enter directory path (or 'prev' to go back, 'pwd' to display, 'exit' to quit): abc
Error: Path does not exist.

Enter directory path (or 'prev' to go back, 'pwd' to display, 'exit' to quit): pwd
Current Directory: /home/bhardwaj/2520090182_OSSP/2520090182_Skill/Skill-09

Enter directory path (or 'prev' to go back, 'pwd' to display, 'exit' to quit): exit
Exiting navigation...
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-09$ |
```

PART 2: Built-in Command Dispatcher

Code:

```
GNU nano 2.9.2 skill9_dispatch.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>

#define MAX_INPUT 100

void cmd_pwd() {
    char cwd[PATH_MAX];

    if (getcwd(cwd, sizeof(cwd)) != NULL)
        printf("Current Directory: %s\n", cwd);
    else
        perror("pwd");
}

void cmd_cd(char *path) {
    if (path == NULL) {
        printf("cd: missing operand\n");
        return;
    }

    if (chdir(path) != 0)
        perror("cd");
    else
        printf("Directory changed successfully.\n");
}

void cmd_help() {
    printf("\nBuilt-in Commands:\n");
    printf("pwd - Display current directory\n");
    printf("cd - Change directory\n");
    printf("help - Display available commands\n");
    printf("exit - Exit the dispatcher\n");
}

void cmd_exit() {
    printf("Exiting command dispatcher...\n");
}
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-09$ ./skill9_dispatch
=== Built-in Command Dispatcher ===

dispatch> help

Built-in Commands:
  pwd  - Display current directory
  cd   - Change directory
  help - Display available commands
  exit - Exit the dispatcher

dispatch> pwd
Current Directory: /home/bhardwaj/2520090182_OSSP/2520090182_Skill/Skill-09

dispatch> cd ..
Directory changed successfully.

dispatch> pwd
Current Directory: /home/bhardwaj/2520090182_OSSP/2520090182_Skill

dispatch> abc
Invalid command: abc
Type 'help' to see available commands.

dispatch> cd OS_PROJECT
cd: No such file or directory

dispatch> pwd
Current Directory: /home/bhardwaj/2520090182_OSSP/2520090182_Skill

dispatch> exit
Exiting command dispatcher...
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-09$ |
```